

A STATISTICAL ANALYSIS OF DENGUE INCIDENCE USING ENVIRONMENTAL AND GEOGRAPHIC FACTORS IN SRI LANKA

ST 3011 - STATISTICAL PROGRAMMING



GROUP 01

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INTRODUCTION

Dengue fever is a major public health concern in Sri Lanka, and predicting the number of dengue cases can support early preventive actions and effective planning. For our study, dengue case data were taken from the National Dengue Control Unit, while location information (latitude, longitude, elevation) came from a Kaggle dataset, and weather data were collected using the Open Meteo Historical Weather API.

Our analysis includes descriptive statistics, graphs, tables, hypothesis testing, and multiple regression models. These methods allow us to explore patterns in the data and identify important factors that may influence dengue cases. The results will provide useful insights for understanding the relationship between environmental conditions and dengue incidence in Sri Lanka.

PROBLEM STATEMENT

The core issue this study seeks to address is the lack of integrated approaches that combine health, geographical, and weather data to predict dengue incidence at the regional level. To bridge this gap, we employ both statistical analysis and predictive modeling techniques, enabling a more comprehensive understanding of the factors driving dengue transmission. Without such integrated models, health authorities struggle to anticipate outbreaks and allocate resources effectively.

OBJECTIVES

- To classify geographic areas into low, medium, and high-risk dengue zones using dengue incidence data.
- To investigate whether elevation influences the variability of dengue cases across districts, and whether low-lying areas show greater fluctuations in dengue incidence compared to high-altitude areas.
- To measure how weather variables (e.g., temperature, rainfall, humidity) are correlated with dengue incidence.
- To study dengue incidence using a Multiple Linear Regression model that combines climatic variables (humidity and temperature), geospatial attributes (longitude, latitude, and elevation).

METHODOLOGY

- A quantitative, data-driven approach was adopted to analyse dengue incidence in relation to environmental, geographic, and temporal factors.
- The study followed a structured analytical workflow consisting of exploratory analysis, statistical testing, and modelling to address the stated objectives.
- Exploratory Data Analysis (EDA) was first conducted to understand data distributions, identify outliers, examine variability, and explore preliminary relationships among variables using summary statistics and visualisations.
- Where appropriate, data were grouped or transformed (e.g., log transformation) to stabilise variance, reduce skewness, and improve interpretability of results.
- Both parametric and non-parametric statistical methods were applied depending on data distribution and underlying assumptions.
- Comparative statistical tests were used to assess differences in dengue incidence and variability across predefined geographic or environmental categories.
- All analyses were implemented using Python, with widely used scientific libraries to ensure reproducibility and transparency.

DATASET

The dataset used for this analysis consists of 900 observations across 11 variables, combining dengue case records (2019–2021) with location and weather information from multiple sources.

- Year: The calendar year in which dengue case data were recorded (2019–2021).
- Province: The administrative province of Sri Lanka where the data point belongs.
- District: The specific district within the province where cases were reported.
- Latitude: Geographic coordinate (north–south position) of the district center.
- Longitude: Geographic coordinate (east–west position) of the district center.
- Elevation: Height above sea level (in meters) of the district location.
- Month: The month of the year (1–12) when dengue cases were recorded.
- Cases: The number of dengue cases reported in that district for the given month.
- Temp_avg: Average temperature (°C) for the district during the month.
- Precipitation_avg: Average rainfall (mm/day equivalent) for the district during the month.
- Humidity_avg: Average relative humidity (%) for the district during the month.

DATA PREPROCESSING

During data preprocessing, missing values were found in one record. This record was removed because it represents a very small part of the dataset, and removing it does not affect the overall distribution or results of the analysis.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 900 entries, 0 to 899
Data columns (total 11 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Year                  900 non-null   int64
1   Province              900 non-null   object
2   District              900 non-null   object
3   Latitude              900 non-null   float64
4   Longitude             900 non-null   float64
5   Elevation             900 non-null   int64
6   Month                 900 non-null   int64
7   Cases                 900 non-null   int64
8   Temp_avg              899 non-null   float64
9   Precipitation_avg     899 non-null   float64
10  Humidity_avg          899 non-null   float64
dtypes: float64(5), int64(4), object(2)
memory usage: 77.5+ KB
```

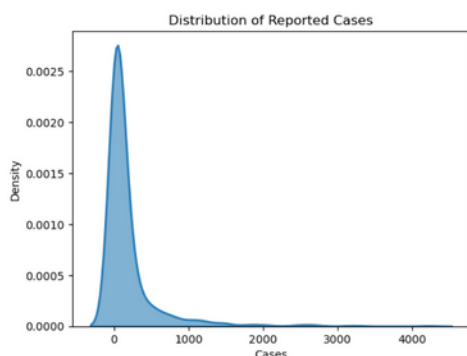
```
<class 'pandas.core.frame.DataFrame'>
Index: 899 entries, 0 to 899
Data columns (total 11 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Year                  899 non-null   int64
1   Province              899 non-null   object
2   District              899 non-null   object
3   Latitude              899 non-null   float64
4   Longitude             899 non-null   float64
5   Elevation             899 non-null   int64
6   Month                 899 non-null   int64
7   Cases                 899 non-null   int64
8   Temp_avg              899 non-null   float64
9   Precipitation_avg     899 non-null   float64
10  Humidity_avg          899 non-null   float64
dtypes: float64(5), int64(4), object(2)
```

Initially, the dataset contained 900 records, and after removing one record with missing data, 899 records remained for analysis.

LITERATURE REVIEW

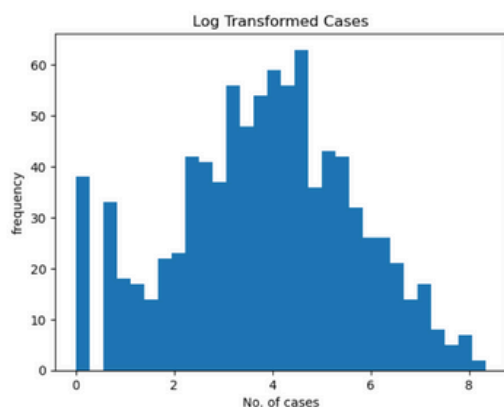
- Research on dengue in Sri Lanka has employed diverse statistical methodologies to understand disease patterns and predict outbreaks. A 2022 study done by Arachchi and Peiris utilized time-series analysis through ARIMA modeling to forecast weekly dengue incidence in Colombo District from 2015 to 2020, demonstrating the efficiency of autoregressive models in epidemiological forecasting. <https://rda.sliit.lk/entities/publication/b54dd389-c1c7-437c-a5be-13556a5d4f55>
- A 2019 investigation applied both Poisson and Negative Binomial regression models to analyze monthly dengue cases from 2010 to 2018, determining that the Negative Binomial model better accommodated overdispersion in count data, with rainfall emerging as a critical predictor including lag effects. www.ijsrp.org/research-paper-0219/ijsrp-p8616.pdf
- Wimalarathna and Jahufer (2023) further contributed by employing EGARCH modeling with lagged climatic predictors to analyze dengue patterns in Gampaha District, achieving substantial explanatory power with R^2 values approaching 0.72. <https://dspace.lib.seu.ac.lk/items/59f8753c-76bb-4496-9ba9-36973c5dbe36>
- A comprehensive systematic review conducted in 2025 examined global forecasting models across endemic regions, synthesizing evidence from institutions including Diponegoro University, Mahidol University, and Universiti Sains Malaysia. The review demonstrated that machine learning algorithms, particularly Random Forest and Long Short-Term Memory networks, frequently outperform conventional time-series models such as ARIMA and SARIMA in predictive accuracy as measured by RMSE, MAE, and MAPE metrics. https://microbiol.crie.ru/jour/article/view/18837/en_US

EXPLORATORY DATA ANALYSIS



The distribution of reported dengue case counts reveals a highly right-skewed distribution. A large proportion of observations are concentrated at low case numbers, while a small number of time periods exhibit extremely high case counts, representing outbreak events. This long right tail indicates substantial variability and the presence of extreme

values that can disproportionately influence statistical analyses. Such skewness violates the normality assumption required for many parametric methods and suggests heteroscedasticity where variability increases with the magnitude of cases. Therefore, the raw case data are not well suited for direct use in correlation or linear regression analysis without transformation.

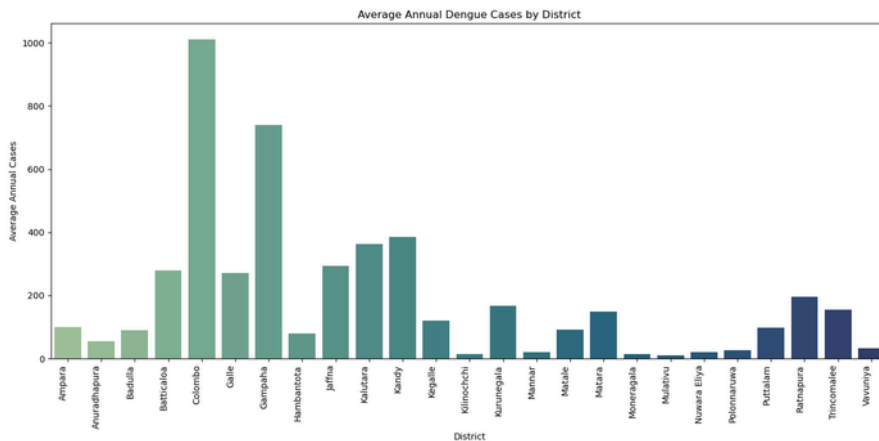


The histogram of log-transformed dengue case counts shows a markedly improved distribution compared to the original data. The transformation reduces the influence of extreme outbreak values and compresses the long right tail, resulting in a distribution that is more symmetric and closer to normality. Variability across observations appears more stable, indicating improved homoscedasticity.

This transformation enhances the suitability of the data for parametric statistical techniques such as correlation analysis and linear regression, as it better satisfies underlying model assumptions. Consequently, the $\log(\text{cases} + 1)$ transformation was adopted for subsequent analyses involving continuous modeling of dengue incidence.

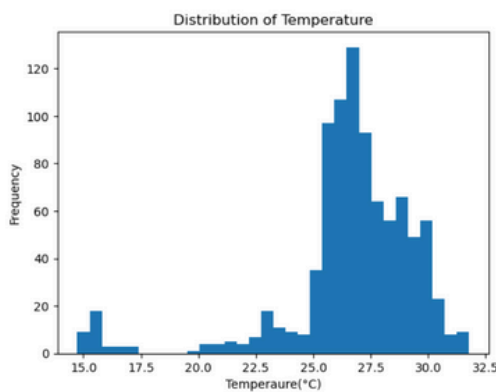
	District	Count		Year	Count
20	Polonnaruwa	36			
21	Trincomalee	36	0	2020	300
22	Ratnapura	36	1	2021	300
23	Vavuniya	36			
24	Gampaha	35	2	2019	299

The dataset contains 36 records for each district, except for the Gampaha district, which has 35 records (partial district output). The data cover three years—2019, 2020, and 2021. While the years 2020 and 2021 each contain 300 records, the year 2019 has only 299 records.

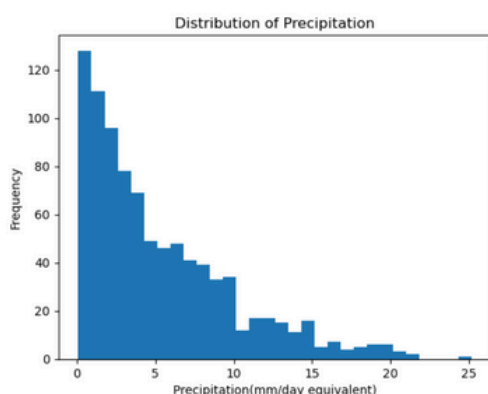


This chart shows the geographic distribution of dengue cases across districts, with Colombo experiencing by far the highest burden at approximately 1,000 average annual cases. Gampaha ranks second with around 750 cases, followed by Kandy and

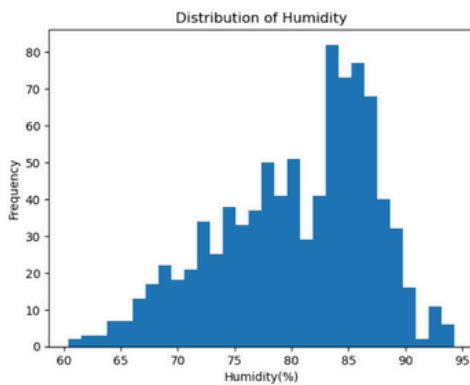
Kalutara with roughly 380-390 cases each. The data reveals a clear urban concentration pattern, as most districts report fewer than 200 cases annually, with some recording less than 25.



The temperature distribution appears negatively skewed with a concentration of values between approximately 25°C and 30°C. This indicates that the study area predominantly experiences warm climatic conditions, which are generally favorable for mosquito activity. A small number of observations occur at lower temperatures, while extremely high temperatures are less frequent. The relatively narrow temperature range may partly explain why dengue transmission appears sensitive to changes within this optimal window rather than across extreme temperature differences.



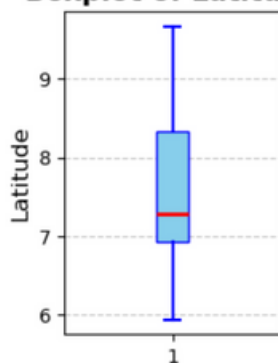
The histogram of precipitation reveals a positively skewed distribution, with most observations concentrated at lower rainfall levels and fewer instances of heavy precipitation. This indicates that light to moderate rainfall conditions dominate the study period, while extreme rainfall events are relatively rare. Such skewness suggests that precipitation data may not follow a normal distribution and could influence statistical modeling if not appropriately handled. The limited frequency of very high rainfall values may also contribute to the weak direct association observed between precipitation and dengue incidence.



The histogram of humidity shows a distribution centered around high humidity levels, generally between 75% and 90%. This suggests that the environment is consistently humid, creating favorable conditions for mosquito survival. The distribution is slightly skewed, with fewer observations at very low humidity levels. The prevalence of high humidity may help explain

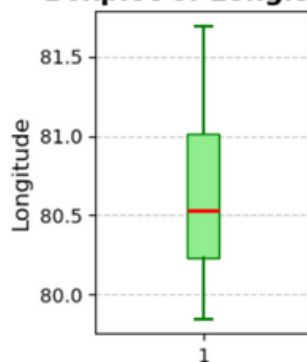
the positive association observed between humidity and dengue incidence, as sustained moisture levels support longer mosquito lifespan and increased virus transmission potential.

Boxplot of Latitude

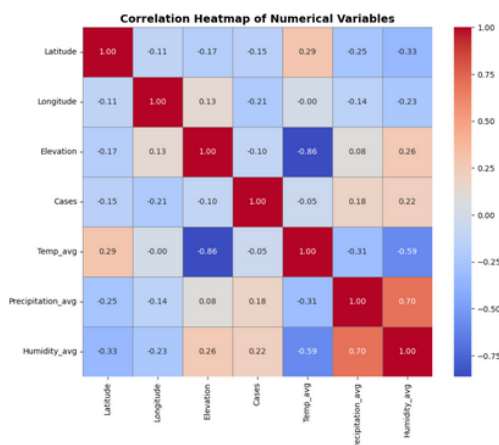


The boxplot of latitude shows a moderately spread distribution ranging from approximately 6.0 to 9.7. The median latitude is around 7.3, with the interquartile range spanning from about 7.0 (Q1) to 8.2 (Q3), indicating a slight right skew in the data.

Boxplot of Longitude



The boxplot of longitude reveals a relatively narrow distribution ranging from approximately 80.0 to 81.5. The median longitude is around 80.5, with the interquartile range spanning from about 80.23 (Q1) to 81.015 (Q3), indicating a fairly symmetric spread of values.



The heatmap shows strong negative correlation between elevation and temperature ($r = -0.86$), and strong positive correlation between precipitation and humidity ($r = 0.70$). Other variables show weak associations with cases

I) CLASSIFICATION OF GEOGRAPHIC AREAS INTO LOW, MEDIUM, AND HIGH-RISK DENGUE ZONES USING DENGUE INCIDENCE DATA.

The dengue surveillance dataset contained three missing values occurring within a single observation corresponding to Gampaha district in March 2019. As the missing values were confined to one row, this observation was excluded from the analysis. To reduce the influence of short-term fluctuations and missing monthly data, monthly dengue case counts were not directly summed. Instead, average monthly dengue cases were calculated for each district for the years 2019, 2020, and 2021.

	District	Year	Cases
0	Ampara	2019	159.000000
1	Ampara	2020	108.083333
2	Ampara	2021	35.166667
3	Anuradhapura	2019	97.000000
4	Anuradhapura	2020	36.083333
5	Anuradhapura	2021	31.000000
6	Badulla	2019	160.166667
7	Badulla	2020	43.750000
8	Badulla	2021	63.583333



	District	Cases
0	Ampara	100.750000
1	Anuradhapura	54.694444
2	Badulla	89.166667
3	Batticaloa	280.000000
4	Colombo	1010.444444

District-level estimates were subsequently aggregated by summing across districts belonging to the same province to derive province-level dengue case estimates.

These yearly averages were then averaged across the three-year period to obtain a single representative monthly dengue case value for each district.

	District	Cases	Province
0	Ampara	100.750000	Eastern
1	Anuradhapura	54.694444	North central
2	Badulla	89.166667	Uva
3	Batticaloa	280.000000	Eastern
4	Colombo	1010.444444	Western

	Province	total_cases
0	Central	497.638889
1	Eastern	535.361111
2	North Western	264.583333
3	North central	80.694444
4	Northern	373.111111
5	Sabaragamuwa	316.055556
6	Southern	499.888889
7	Uva	103.972222
8	Western	2111.505051

Provincial population data were obtained from the Central Bank of Sri Lanka, enabling adjustment of dengue case estimates relative to population size. Province-wise proportions and population-adjusted dengue incidence were then calculated.

(https://www.cbsl.gov.lk/sites/default/files/cbslweb_documents/statistics/otherpub/ess_2020_e1.pdf)

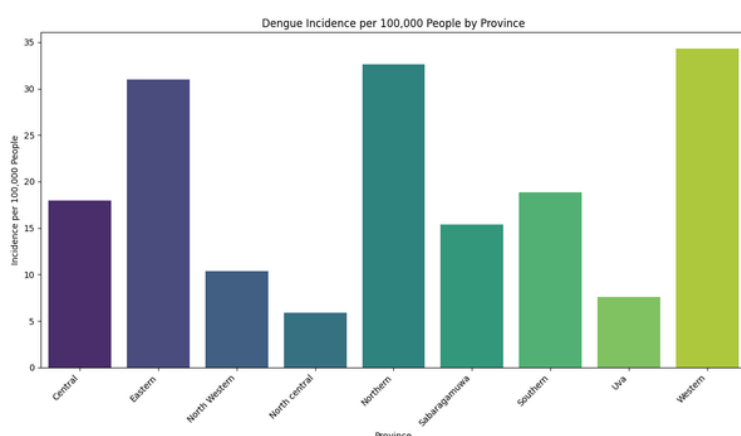
Province	Mid Year Population (2018) '000	Land Area (Sq.km.)	Density of Population
Western	6,149	3,593	1,711
Central	2,766	5,575	496
Southern	2,654	5,383	493
Northern	1,143	8,290	138
Eastern	1,729	9,361	185
North Western	2,551	7,506	340
North Central	1,377	9,741	141
Uva	1,376	8,335	165
Sabaragamuwa	2,058	4,921	418
Total	21,803	62,705	348

Composition of Population (Census of Population and Housing – 2012)

	Province	incidence_per_100k	risk_level
0	Central	17.991283	Medium Risk
1	Eastern	30.963627	High Risk
2	North Western	10.371750	Low Risk
3	North central	5.860163	Low Risk
4	Northern	32.643142	High Risk
5	Sabaragamuwa	15.357413	Medium Risk
6	Southern	18.835301	Medium Risk
7	Uva	7.556121	Low Risk
8	Western	34.338999	High Risk

Provinces were classified into low-, medium-, and high-risk dengue zones using percentile-based quantile thresholds.

Graphical Analysis



The bar chart indicates that dengue incidence varies considerably by province, with the Western Province reporting the highest cases per 100,000 people, followed closely by the Northern and Eastern Provinces. Southern and Central Provinces show moderate incidence levels, while Sabaragamuwa is slightly lower. North Central Province has the lowest dengue incidence overall.

Three provinces (Western, Northern, and Eastern) account for the highest disease burden with incidence rates exceeding 30 per 100,000 population, while three provinces (North Central, Uva, and North Western) maintain rates below 11 per 100,000. This indicates critical geographic inequalities requiring targeted public health interventions in high-risk regions.

High risk provinces

- Eastern
- Northern
- Western

Medium risk provinces

- Central
- Sabaragamuwa
- Southern

Low risk provinces

- North Central
- North Western
- Uva

II) EFFECT OF ELEVATION ON THE VARIABILITY OF DENGUE CASES

Data Grouping by Elevation

Districts were categorized into three elevation bands as Low, Medium, and High elevation using elevation quantiles to ensure balanced group sizes.

```
Elevation_Group
Low Elevation      360
High Elevation     288
Medium Elevation   252
Name: count, dtype: int64
```

This grouping allows for a fair comparison of dengue case variability across different elevation levels.

Descriptive Analysis of Dengue Case Variability

To assess fluctuations in dengue incidence, descriptive statistics focusing on variability were computed for each elevation group. These included the mean, variance, standard deviation, and coefficient of variation (CV) of dengue cases.

	Mean	Variance	Std_Dev	Coeff_of_Variation
Elevation_Group				
Low Elevation	244.961111	249436.850851	499.436533	2.038840
Medium Elevation	197.805556	164913.352479	406.095251	2.053002
High Elevation	115.274306	54110.617874	232.616891	2.017942

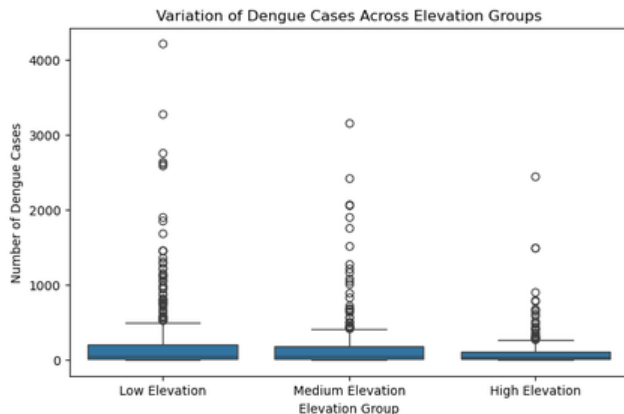
Interpretation

- Low-elevation districts exhibit the highest absolute variability, as indicated by the largest variance and standard deviation.
- While the coefficient of variation is high across all groups ($CV \approx 2$), low-elevation areas show both higher average dengue cases and larger fluctuations in absolute terms.
- High-elevation districts display substantially lower variance, suggesting more stable dengue incidence patterns.

This supports the hypothesis that dengue incidence is more unstable in low-lying areas.

Visualization of Dengue Case Variability

A boxplot was used to visually compare the distribution of dengue cases across elevation groups.



Key observations from the boxplot:

- Low-elevation districts show a wider spread and a greater number of extreme outliers, representing sudden dengue outbreaks.
 - High-elevation districts display a narrower interquartile range, indicating relatively stable dengue case counts.
- The presence of numerous upper outliers across all groups reflects the outbreak-driven nature of dengue incidence.

The visual evidence is consistent with the descriptive statistics and highlights greater fluctuations in low-elevation regions.

Statistical Test for Equality of Variances

To formally test whether dengue case variability differs across elevation groups, **Levene's Test** for Equality of Variances was applied.

- Test statistic: 8.352
- p-value: 0.000255

Hypotheses

- H_0 : Dengue case variance is equal across elevation groups
- H_1 : At least one elevation group has a different variance

Interpretation

Since the p-value is less than 0.05, the null hypothesis is rejected. This indicates that dengue case variability differs significantly across elevation levels.

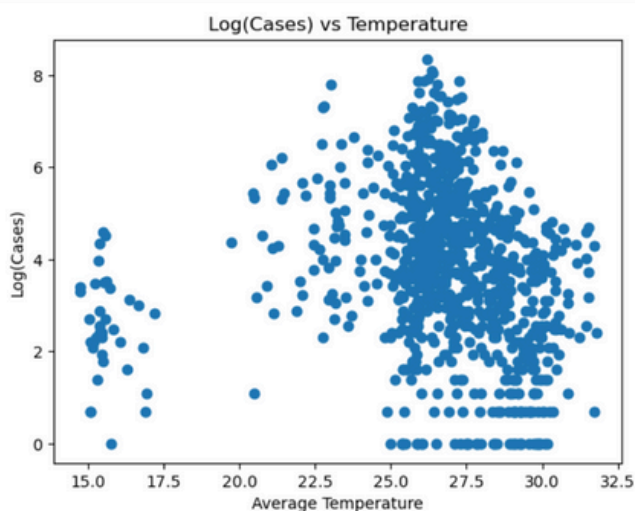
The analysis provides strong statistical evidence that elevation influences the variability of dengue cases across districts. **Low-elevation areas experience greater fluctuations and more extreme outbreaks, while high-altitude regions exhibit lower variability and more stable dengue incidence.**

III) HOW WEATHER VARIABLES ARE CORRELATED WITH DENGUE CASES

Dengue fever is a major public health concern in many tropical and subtropical regions, where climatic conditions play a crucial role in influencing the transmission dynamics of the disease. Weather variables such as temperature, rainfall, and humidity affect the breeding, survival, and activity of *Aedes* mosquitoes, the primary vectors of dengue. Variations in these environmental factors can therefore lead to fluctuations in dengue incidence over time. Understanding the relationship between weather conditions and dengue cases is essential for developing effective early warning systems and targeted vector control strategies.

In this statistical analysis, the objective is to measure and quantify the correlation between key weather variables—namely temperature, rainfall, and humidity—and dengue incidence, using appropriate exploratory and inferential statistical methods. The findings aim to provide evidence-based insights into the extent to which climatic factors are associated with dengue transmission patterns.

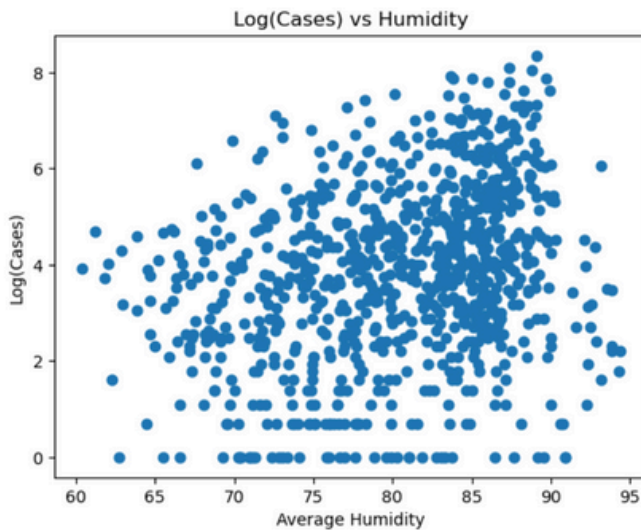
Log(Cases) vs Temperature



The scatter plot of log-transformed dengue cases versus average temperature suggests a nonlinear relationship. Higher case levels are primarily clustered within a moderate temperature range of approximately 25°C to 29°C, indicating that dengue transmission may be more intense within this optimal temperature window. At lower temperatures, dengue incidence appears consistently low, while at very high temperatures the variability increases

but does not show a clear upward trend. This pattern aligns with biological evidence that mosquito survival and viral replication occur most efficiently within a specific temperature range. Despite this, the spread of points indicates that temperature alone does not fully explain fluctuations in dengue cases.

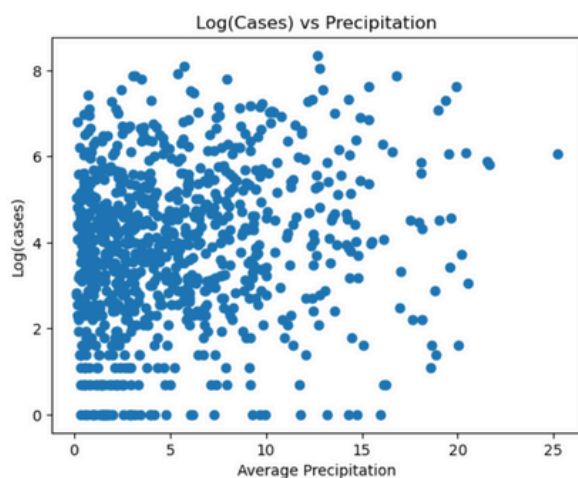
Log(Cases) vs Humidity



The scatter plot of log-transformed dengue cases against average humidity demonstrates a clearer positive association compared to other weather variables. Higher dengue incidence tends to occur at elevated humidity levels, particularly above approximately 75–80%. The clustering of higher log(case) values in this humidity range suggests that increased moisture in the environment supports mosquito survival and enhances transmission potential. However, the

presence of variability across the humidity spectrum indicates that additional environmental or temporal factors also contribute to dengue outbreaks.

Log(Cases) vs Precipitation



The scatter plot of log-transformed dengue cases against average precipitation indicates a weak and highly variable relationship. While moderate increases in log(cases) are observed across a range of precipitation levels, the data points are widely dispersed, and no strong linear trend is evident. High dengue incidence occurs at both low and moderate precipitation levels, suggesting that rainfall alone may not directly determine case numbers.

This pattern may reflect the complex role of precipitation in mosquito ecology, where moderate rainfall can create breeding sites, while excessive rainfall may disrupt larval habitats. The variability in the plot suggests that precipitation likely influences dengue transmission indirectly and possibly with time-lagged effects.

Spearman's rank Correlation

Because the scatter plots did not show a clear linear relationship and the data exhibited considerable spread, Spearman's rank correlation was used to assess the association between weather variables and log-transformed dengue cases.

Temperature

Hypotheses

- H_0 : No correlation between temperature and Log (Cases)
- H_1 : There is a significant correlation

Spearman correlation (Temperature vs Log Cases): -0.2526

p-value: 1.4739857153060816e-14

Interpretation

A statistically significant negative correlation was found between temperature and dengue incidence (Spearman's $\rho = -0.253$, $p < 0.001$), suggesting that higher temperatures may be associated with reduced case counts within the observed range.

Humidity

Hypotheses

- H_0 : No correlation between Humidity and Log (Cases)
- H_1 : There is a significant correlation

Spearman correlation (Humidity vs Log Cases): 0.2755

p-value: 4.018995868465933e-17

Interpretation

$|0.2755|$ is less than 0.3. it implies that there's a very weak correlation between humidity and log(cases).

Precipitation

Hypotheses

- H_0 : No correlation between precipitation and Log (Cases)
- H_1 : There is a significant correlation

Pearson correlation (Precipitation vs Log Cases): 0.1668

p-value: 4.905606529108394e-07

Interpretation

A weak but statistically significant positive correlation was observed between precipitation and dengue incidence.

IV) STATISTICAL MODEL FITTING AND INSIGHTS

A statistical model is required to quantify relationships between variables and make reliable predictions from data.

One of objectives of this study was to examine dengue incidence using a Multiple Linear Regression model that combines climatic variables (humidity and temperature) with geospatial attributes (longitude, latitude, and elevation). In line with this objective, Year was included as a categorical variable to account for annual differences, while Month was excluded to focus on broader inter-annual variation rather than seasonal cycles.

Forward selection was applied to identify the variables most strongly associated with the response variable, dengue cases. The Multiple Linear Regression model obtained from the analysis, using only base terms without log transformations or quadratic effects, is given below.

Model Equation (Python Output):

Cases_predicted = 12586.770 + -239.962*C(Year)[T.2020] + -258.878*C(Year)[T.2021] + 1.801*Humidity_avg + -129.925*Longitude + -0.479*Elevation + -54.160*Temp_avg + -49.893*Latitude

Model Summary (Python Output):

OLS Regression Results

Dep. Variable:	Cases	R-squared:	0.207
Model:	OLS	Adj. R-squared:	0.201
Method:	Least Squares	F-statistic:	33.33
Date:	Thu, 04 Dec 2025	Prob (F-statistic):	2.58e-41
Time:	18:33:09	Log-Likelihood:	-6573.5
No. Observations:	899	AIC:	1.316e+04
Df Residuals:	891	BIC:	1.320e+04
Df Model:	7		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
Intercept	1.259e+04	2141.845	5.877	0.000	8383.121	1.68e+04
C(Year)[T.2020]	-239.9618	29.794	-8.054	0.000	-298.436	-181.488
C(Year)[T.2021]	-258.8776	29.997	-8.630	0.000	-317.751	-200.004
Humidity_avg	1.8013	2.941	0.612	0.540	-3.971	7.573
Longitude	-129.9250	24.981	-5.201	0.000	-178.954	-80.896
Elevation	-0.4794	0.081	-5.917	0.000	-0.638	-0.320
Temp_avg	-54.1598	12.524	-4.325	0.000	-78.740	-29.580
Latitude	-49.8932	12.878	-3.874	0.000	-75.168	-24.619

Omnibus:	826.686	Durbin-Watson:	1.182
Prob(Omnibus):	0.000	Jarque-Bera (JB):	27758.787
Skew:	4.211	Prob(JB):	0.00
Kurtosis:	28.887	Cond. No.	7.47e+04

The fitted model yielded a coefficient of determination (R^2) of 20.7%, indicating that the regression explains 20.7% of the variance in dengue cases. This relatively low explanatory power suggests that the model could be improved by incorporating additional predictors or alternative modeling approaches.

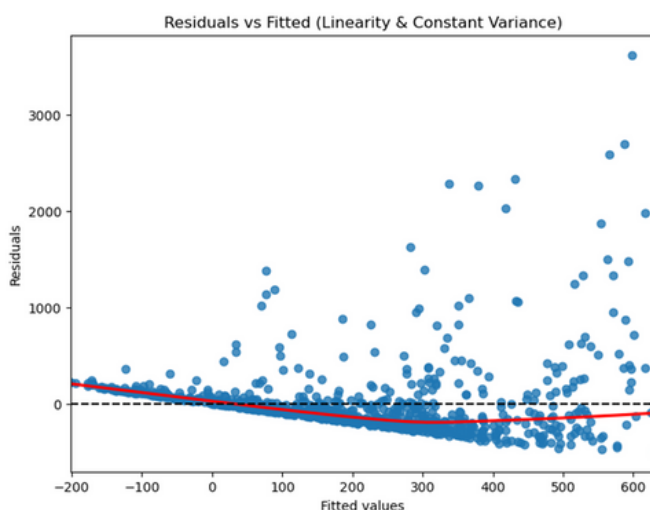
H_0 : Regression model is not significant.

H_1 : Regression model is significant.

At the 5% significance level, the p-value associated with the F-statistic ($2.58e-41$) is far less than 0.05. Therefore, we reject the null hypothesis (all regression coefficients are equal to zero) and conclude that the regression model exhibits overall statistical significance at the 5% level.

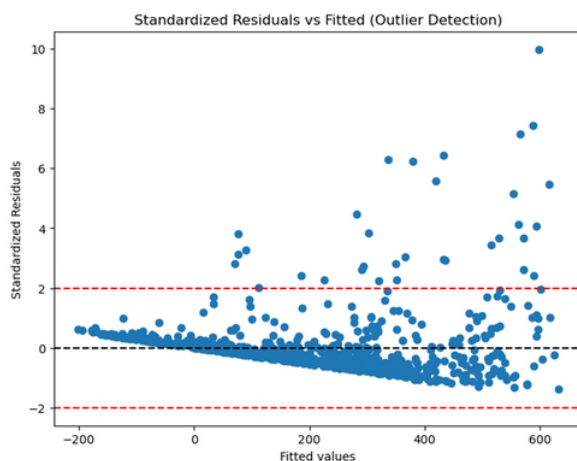
DIAGNOSTIC PLOTS

Diagnostic plot for Residuals Vs Predicted values



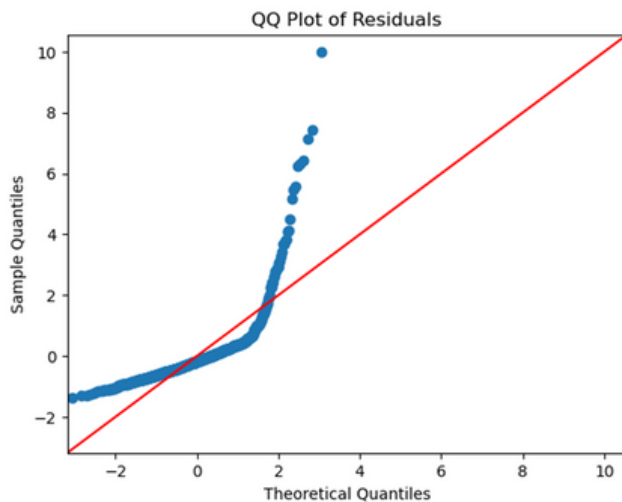
The residuals versus predicted values plot indicates that the data points do not follow a linear pattern. A clear curvilinear trend is visible, suggesting that the linearity assumption may not hold. Furthermore, the wedge-shaped spread of residuals implies heteroscedasticity, meaning the assumption of constant variance is violated. These patterns highlight potential inadequacies in the model

specification and suggest that alternative predictors or transformations may be required to improve model fit.



Several residuals are plotted beyond ± 2 on the standardized residual scale, which suggests the presence of regression outliers.

Normal probability plot for residuals.



The QQ plot was used to assess the assumption of normality of residuals. The plot shows noticeable deviations from the straight reference line, indicating that the residuals are not normally distributed and that the normality assumption is violated.

Therefore, we aim to modify the model to better satisfy the standard regression assumptions and to improve the coefficient of determination.

IMPROVED MODEL

To address violations of regression assumptions, we applied a log transformation to stabilize non-constant variance and introduced second-order (quadratic) terms for numerical predictors to better capture non-linear relationships. These modifications were undertaken not only to improve compliance with regression assumptions but also to enhance the explanatory power of the model, as reflected in the coefficient of determination (R^2).

Model equation (python output):

```
log(Cases)_predicted = 9060.179 + -1.385*C(Year)[T.2020] + -1.630*C(Year)[T.2021]  
+ -0.311*Humidity_avg + 0.002*I(Humidity_avg ** 2) + -223.678*Longitude +  
1.382*I(Longitude ** 2) + -0.002*Elevation + -0.000*I(Elevation ** 2) +  
0.441*Temp_avg + -0.016*I(Temp_avg ** 2) + 2.415*Latitude + -0.197*I(Latitude ** 2)
```

Model summary (python output):

```

=====
                    OLS Regression Results
=====
Dep. Variable:      np.log(Cases)    R-squared:          0.537
Model:              OLS              Adj. R-squared:     0.530
Method:             Least Squares    F-statistic:        81.97
Date:               Thu, 04 Dec 2025  Prob (F-statistic):    8.46e-133
Time:               20:16:52          Log-Likelihood:     -1381.3
No. Observations:   861              AIC:                2789.
Df Residuals:       848              BIC:                2850.
Df Model:           12
Covariance Type:    nonrobust
=====
               coef    std err          t      P>|t|      [0.025    0.975]
-----
Intercept      9060.1790    1181.728      7.667      0.000     6740.723    1.14e+04
C(Year)[T.2020] -1.3848      0.101    -13.675      0.000     -1.584     -1.186
C(Year)[T.2021] -1.6300      0.102    -15.925      0.000     -1.831     -1.429
Humidity_avg    -0.3113      0.152     -2.047      0.041     -0.610     -0.013
I(Humidity_avg ** 2) 0.0018      0.001      1.864      0.063     -9.51e-05     0.004
Longitude      -223.6777     29.246     -7.648      0.000    -281.082    -166.274
I(Longitude ** 2)   1.3823      0.181      7.632      0.000      1.027      1.738
Elevation       -0.0018      0.001     -3.255      0.001     -0.003     -0.001
I(Elevation ** 2)  -3.483e-07    3.81e-07    -0.914      0.361    -1.1e-06    3.99e-07
Temp_avg        0.4406      0.474      0.929      0.353     -0.490      1.371
I(Temp_avg ** 2)   -0.0163      0.009     -1.863      0.063     -0.033     0.001
Latitude        2.4147      0.727      3.320      0.001      0.987      3.842
I(Latitude ** 2)   -0.1965      0.047     -4.178      0.000     -0.289     -0.104
=====
Omnibus:          6.475    Durbin-Watson:      1.393
Prob(Omnibus):    0.039    Jarque-Bera (JB):    8.345
Skew:             0.057    Prob(JB):            0.0154
Kurtosis:         3.469    Cond. No.            2.03e+10
=====

```

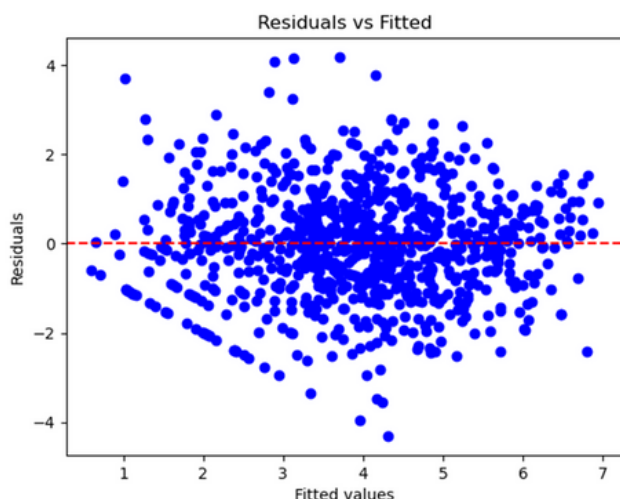
Notes:

In the initial model, the coefficient of determination (R^2) was 20.7%, meaning the predictors explained only about one-fifth of the variation in the response variable. After applying a log transformation to the dependent variable and introducing quadratic terms for the numerical predictors, the R^2 increased substantially to 53.7%.

This improvement indicates that the modified model captures the underlying structure of the data much more effectively. The log transformation helped stabilize non-constant variance and improve the fit, while the quadratic terms allowed the model to account for non-linear relationships between predictors and the response. Together, these adjustments enhanced both compliance with regression assumptions and the explanatory power of the model.

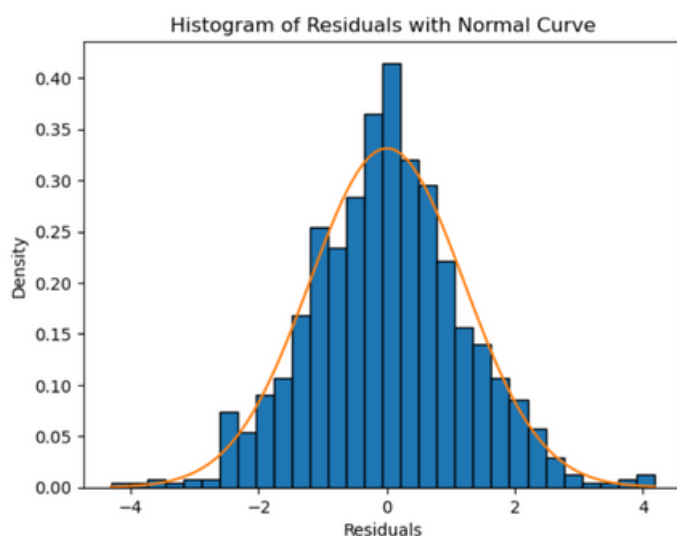
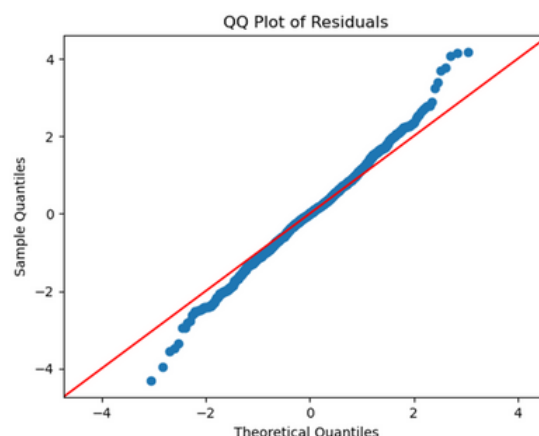
Diagnostic plots were used to assess whether the standard regression assumptions were satisfied in the improved model.

Diagnostic plot for Residuals Vs Predicted values



The residuals versus fitted values plot shows no evidence of a wedge-shaped or curvilinear pattern. The points appear randomly scattered around the zero line, indicating that the assumptions of linearity and constant variance are reasonably satisfied after the model modifications.

The QQ plot of residuals from the improved model shows that the data points closely align with the reference line, indicating that the residuals are approximately normally distributed. This suggests that the normality assumption is reasonably satisfied, enhancing the reliability of statistical inference based on the regression model.



Shapiro-Wilk Test Statistic: 0.9963

The residual histogram of the final log-transformed model shows an approximately normal distribution. This is supported by the Shapiro-Wilk test statistic ($W = 0.9963$), indicating near-normal residuals. Along with the improved variance pattern, the main regression assumptions are considered to be reasonably satisfied.

OVERALL INTERPRETATION OF THE FINAL MODEL FOR DENGUE INCIDENCE

The final model indicates that dengue cases (on the log scale) are influenced by temporal effects (Year), climatic factors (humidity and temperature), and geographical characteristics (latitude, longitude, and elevation). The inclusion of quadratic terms shows that several predictors have non-linear effects on dengue incidence.

CONCLUSION

The initial regression model showed statistical significance ($p < 0.05$) but explained only 20.7% of dengue case variation and violated key assumptions like linearity, constant variance, and normality. After applying a log transformation and adding quadratic terms, the improved model achieved a higher R^2 of 53.7%, with diagnostic plots confirming that assumptions were reasonably satisfied. Residuals appeared normally distributed (Shapiro-Wilk $W = 0.9963$), and variance patterns improved, making the final model statistically sound and better suited for capturing both linear and non-linear effects of climatic and geospatial predictors.

GENERAL CONCLUSIONS

- Based on the exploratory data analysis, dengue transmission is primarily influenced by warm temperatures (25-30°C) and high humidity (75-90%), with most cases occurring during light to moderate rainfall conditions. The disease shows a strong urban concentration pattern, particularly in Colombo and Gampaha districts, and appears sensitive to elevation and temperature variations.
- Dengue risk varies by province with Western, Northern, and Eastern provinces showing high risk indicating a greater need for intensified surveillance and control measures. In contrast, Central, Southern, and Sabaragamuwa provinces fall under medium risk, while North Western, North Central, and Uva provinces exhibit lower dengue risk.
- Dengue case variability differs across elevation levels, with low-lying districts showing greater fluctuations and more extreme outbreaks than high-altitude regions. This indicates that elevation is an important geographic factor influencing the instability of dengue transmission and should be considered in elevation-specific surveillance and preparedness strategies.
- Weather variables show significant but weak associations with dengue incidence, indicating climate plays a role but is not the only driver of outbreaks.
- Humidity has the strongest relationship with dengue cases. Higher humidity levels are linked with increased incidence, likely because moist conditions improve mosquito survival and virus transmission.
- Temperature shows a weak negative correlation with dengue cases. Transmission appears to be highest within an optimal moderate temperature range (around 25–29 °C) rather than continuously increasing with higher temperatures.
- Precipitation has only a weak positive association with dengue incidence. Rainfall alone does not directly determine case numbers and may influence transmission indirectly by creating breeding sites under moderate rainfall.
- The initial Multiple Linear Regression model was statistically significant but explained only a small portion of dengue variation and did not meet key regression assumptions. Model improvements (data transformation and non-linear terms) show that climate factors, particularly temperature and rainfall, are strong predictors of dengue cases. This suggests that monitoring weather patterns can help predict and prepare for dengue outbreaks.